# 4

# *The zeros of the zeta-function and the Prime Number Theorem*

## 4.1

In Chapter 3 we have seen the remarkable correspondence that relates the prime numbers with the zeros of the zeta-function. In order to derive significant conclusions concerning the asymptotic distribution of prime numbers it is necessary to describe that of the zeros. This is in its turn possible through the connection with prime numbers, i.e. through the explicit formulae. The basic example of this is given by Hadamard's zero-free region Theorem 4.2. In fact, the argument given here, apart from some minor variations, is the only one available at present. Somewhat better estimates are available but they depend on subsidiary estimates for the zeta-function (see Exercise 4.7); these in their turn are based on estimates for exponential sums of the form $\sum_{N<n\leqslant M} n^{-it}$ and the approximate functional equation. The estimation of the exponential sums involves techniques which belong to different circles of ideas than those forming the subject matter of this book. See Chapter 6 for more details.

## 4.2

**Theorem:** *There exists a constant $c>0$ so that if $\rho\in Z$ then*

$$\mathrm{Re}(\rho) < 1 - c\,\mathrm{Min}(1, \{\log|\mathrm{Im}(\rho)|\}^{-1}).$$

*Proof*: The proof of this theorem is based on an application of Theorem 3.3 and a remarkable but simple inequality. This is

$$3 + 4\cos x + \cos 2x = 2(1+\cos x)^2 \geqslant 0.$$

It follows from this that if $\sigma > 1$ then

$$\mathrm{Re}\{-3\zeta'(\sigma)/\zeta(\sigma) - 4\zeta'(\sigma+it)/\zeta(\sigma+it) - \zeta'(\sigma+2it)/\zeta(\sigma+2it)\} \geqslant 0 \quad (4.1)$$

since the left-hand side is equal to

$$\sum_{k>0,p} (\log p) p^{-k\sigma}\{3 + 4\cos(kt\log p) + \cos(2kt\log p)\}$$

which is a sum of positive terms.

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From Theorem 3.3 we have that for $0 < \mathrm{Re}(s) < 2$, $|\mathrm{Im}(s)| > 1$

$$-\zeta'(s)/\zeta(s) = -\sum_{\rho:|\mathrm{Im}(s-\rho)|<1} (s-\rho)^{-1} + O(\log|s|).$$

If $\mathrm{Re}(s) > 1$ then $\mathrm{Re}(s-\rho) > 0$ and so $\mathrm{Re}\{(s-\rho)^{-1}\} > 0$. Thus in the region $\{s \in \mathbf{C}: 1 < \mathrm{Re}(s) < 2, |\mathrm{Im}(s)| > 1\}$ one has for a certain constant $c > 0$

$$\begin{aligned}\mathrm{Re}\{-\zeta'(s)/\zeta(s)\} &< -\mathrm{Re}\{(s-\rho_1)^{-1}\} + c(1+\log|s|) \\ &< c(1+\log|s|)\end{aligned}$$

where $\rho_1$ is a zero satisfying $|\mathrm{Im}(s-\rho_1)| < 1$. We shall now combine this with the inequality above as follows. Let $\rho$ be a fixed zero of the zeta-function and let $t = \mathrm{Im}(\rho)$. We know that $-\zeta'(\sigma)/\zeta(\sigma) < (\sigma-1)^{-1} + c'$ for $1 < \sigma$ and a certain constant $c'$. Using this and the first inequality for $\mathrm{Re}\{-\zeta'(s)/\zeta(s)\}$ for $\sigma + it$ with $\rho_1 = \rho$ and the second for $\sigma + 2it$ we obtain

$$-3(\sigma-1)^{-1} + 4\{\sigma - \mathrm{Re}(\rho)\}^{-1} + c' + c(2 + 2\log|t| + \log 2) \geqslant 0.$$

Let

$$\sigma = 1 + \{1 - \mathrm{Re}(\rho)\};$$

we then obtain

$$-\{1-\mathrm{Re}(\rho)\}^{-1} + c' + c(2 + 2\log|t| + \log 2) \geqslant 0$$

which, since $t = \mathrm{Im}(\rho)$, gives the estimate of the theorem if $|\mathrm{Im}(\rho)| > 1$.

To complete the proof it suffices to show that there are no zeros with $\mathrm{Re}(\rho) = 1$, $|\mathrm{Im}(\rho)| \leqslant 1$. Suppose there are such zeros and let $\rho$ be such that $|\mathrm{Im}(\rho)|$ is maximal. Then if $\rho = 1 + it$ in view of our assumption and what was proved above there are no zeros $\rho'$ with $\mathrm{Re}(\rho') = 1$, $|\mathrm{Im}(\rho')| > |\mathrm{Im}(\rho)|$. In particular $1 + 2it$ is not a zero. With this value of $t$ we use the inequality (4.1) again and let $\sigma \to 1$. The first term is asymptotic to $3(\sigma-1)^{-1}$, the second to $-4(\sigma-1)^{-1}$ and the third is bounded. Thus the left-hand side would tend to $-\infty$ as $\sigma \to 1$ and this contradicts the inequality. Hence there is no such zero.

## 4.3

Theorem 4.2 forms the basis of the proof of the Prime Number Theorem as we shall show. In fact the Prime Number Theorem is based on the fact that there are no zeros of the Riemann zeta-function on $1 + i\mathbf{R}$ which is much less than we have actually proved. This deduction can be made by using a Tauberian theorem (such as the Wiener–Ikehara Theorem) which, although it gives a painless proof, is not really open to improvements, such as the determination of error terms.

The first version of the Prime Number Theorem is the following:

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**Theorem:** *There exists a constant $c > 0$ so that as $X \to \infty$*

$$\sum_{n<X} \Lambda(n) = X + O(X e^{-c(\log X)^{1/2}}).$$

Although this theorem does not correspond to the naive way that one might think about the Prime Number Theorem, for example, as formulated in Theorem 4.5 below, the form given here is the most useful.

Note that

$$X e^{-c(\log X)^{1/2}} = O(X/(\log X)^N)$$

as $X \to \infty$ for any $N > 0$. On the other hand we can write

$$X e^{-c(\log X)^{1/2}} = X^{1-c/(\log X)^{1/2}}$$

from which it follows that there is no $a < 1$ such that

$$X e^{-c(\log X)^{1/2}} = O(X^a).$$

## 4.4

In this section we shall prove Theorem 4.3. The idea is to use Theorem 3.8 with a suitable choice of $T$. Let us first use Theorem 4.2 to estimate

$$\sum_{\substack{\rho \in Z \\ |\mathrm{Im}(\rho)| < T}} |X^\rho|/\rho.$$

By Theorem 4.2 one has

$$X^\rho \leqslant X^{1-c(\log T)^{-1}}$$

as long as $T > 3$. Moreover by Corollary 3.4

$$\sum_{|\mathrm{Im}(\rho)| < T} |\rho|^{-1} = O\left(\sum_{0<m<T} m^{-1} \log m\right)$$
$$= O((\log T)^2)$$

by comparing the series with the corresponding integral as in Cauchy's test for convergence.

In the formula which is to be proved we may assume that $X$ is such that $X - 1/2$ is an integer since the sum varies by at most $O(\log X)$ (as $\Lambda(n) \leqslant \log n$) as $X$ varies over an interval of length 1 and on the right-hand side both $X$ and $X e^{-c(\log X)^{1/2}}$ vary by a bounded quantity.

With the assumption that $X$ is of this form, Theorem 3.8 and the estimate we have just made yield

$$\sum_{n<X} \Lambda(n) = X + O(X^{1-c/\log T}) + O(1) + O(X(\log X)/T)$$
$$+ O(X(\log T)^2/T) + O(X).$$

We now choose $T$ so that $\log T = (\log X)^{1/2}$ and, as $X = e^{\log X}$ we obtain

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for the right-hand side

$$X + O(Xe^{-c(\log X)^{1/2}}) + O(1) + O(X(\log X)e^{-(\log X)^{1/2}}) + O(\log X).$$

As $\log X = e^{\varepsilon(\log X)^{1/2}}$ for any $\varepsilon > 0$ we see that for any $c_1 < c$ this is

$$X + O(Xe^{-c_1(\log X)^{1/2}}).$$

Using $c_1$ as the constant whose existence is asserted in the statement of the theorem we see that we have now proved the estimate of the theorem.

## 4.5

As a corollary of the theorem which we have just proved we shall deduce an estimate on

$$\pi(X) = \text{Card}\{p: p \text{ prime}, p \leqslant X\}.$$

The form of the Prime Number Theorem which corresponds to the original calculations and conjectures is:

**Theorem**: *There exists a constant $c > 0$ so that*

$$\pi(X) = \int_2^X (\log u)^{-1}\,du + O(Xe^{-c(\log X)^{1/2}}).$$

Before we prove this we observe that by integration by parts one has

$$\int_2^X (\log u)^{-1}\,du = X/\log X - 2/\log 2 + \int_2^X (\log u)^{-2}\,du.$$

Since for $X > 4$

$$\int_2^X (\log u)^{-2}\,du = \int_2^{X^{1/2}} (\log u)^{-2}\,du + \int_{X^{1/2}}^X (\log u)^{-2}\,du < \int_2^{X^{1/2}} (\log 2)^{-2}\,du + \int_{X^{1/2}}^X (\log X^{1/2})^{-2}\,du < 3X^{1/2} + 4X/(\log X)^2.$$

Thus

$$\int_2^X (\log u)^{-2}\,du = O(X/(\log X)^2)$$

and so it follows that

$$\pi(X) = X/(\log X) + O(X/(\log X)^2)$$

and in particular

$$\pi(X) \sim X/\log X.$$

We shall now deduce this form of the Prime Number Theorem from Theorem 4.3 using the basic technique of partial summation.

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First of all we note that

$$\sum_{n<X}\Lambda(n)=\sum_{p,k:p^k<X}\log p$$
$$=\sum_{p<X}\log p+\sum_{p,k\geqslant 2:p^k<X}\log p,$$

where we use the convention that $p$ represents a prime. In the second summation for any given prime $p$ there are at most $\log X/\log p$ summands, and as $k\geqslant 2$, one also has $p\leqslant X^{1/2}$. Thus the second term is bounded by

$$\sum_{2\leqslant p\leqslant X^{1/2}}\log X\leqslant X^{1/2}\log X.$$

It follows that Theorem 4.3 can be phrased as

$$\sum_{p<X}\log p=X+O(X\mathrm{e}^{-c(\log X)^{1/2}}).$$

Denote the left-hand side by $A(X)$. Then

$$\{A(n+1)-A(n)\}/\log n=\begin{matrix}1\\0\end{matrix}\Bigg\}\begin{matrix}\text{if } n \text{ is a prime}\\ \text{if } n \text{ is not a prime.}\end{matrix}$$

Thus

$$\pi(X)=\sum_{n\leqslant X}\{A(n+1)-A(n)\}/\log n$$
$$=\{A(3)-A(2)\}/\log 2+\{A(4)-A(3)\}/\log 3+\cdots$$
$$+\{A([X]+1)-A([X])\}/\log[X].$$

Note that $A(2)=0$ and group the terms $A(3)$, $A(4)$ and so on together; one finds that

$$\pi(X)=A(3)\{(\log 2)^{-2}-(\log 3)^{-1}\}+A(4)\{(\log 3)^{-1}-(\log 4)^{-1}\}+\cdots$$
$$+A([X])\{(\log[X-1])^{-1}-(\log[X])^{-1}\}+A([X]+1)/\log[X].$$

We know that for certain constants $c,c_1>0$ and $j\leqslant X$

$$|A(j)-j|<c_1X\mathrm{e}^{-c(\log X)^{1/2}}.$$

Thus we find that

$$\pi(X)=3\{(\log 2)^{-1}-(\log 3)^{-1}\}+\cdots$$
$$+[X]\{(\log[X-1])^{-1}-(\log[X])^{-1}\}$$
$$+([X]+1)/\log[X]+E(X)$$

where

$$E(X)\leqslant c_1X\mathrm{e}^{-c(\log X)^{1/2}}(\{(\log 2)^{-1}-(\log 3)^{-1}\}+\{(\log 3)^{-1}-(\log 4)^{-1}\}$$
$$+\cdots+\{(\log[X]-1)^{-1}-(\log[X])^{-1}\}+(\log[X])^{-1})$$
$$=c_1X\mathrm{e}^{-c(\log X)^{1/2}}(\log 2)^{-1}.$$

If we now group the terms in $(\log 2)^{-1}$, $(\log 3)^{-1}$ etc. together then we have

$$\pi(X)=3(\log 2)^{-1}+\{(\log 3)^{-1}+(\log 4)^{-1}+\cdots+(\log[X])^{-1}\}+E(X).$$

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By the argument used in the proof of Cauchy's criterion for the convergence of a sum in terms of the corresponding integral we have

$$(\log 2)^{-1} + (\log 3)^{-1} + \cdots + (\log [X])^{-1} < (\log 2)^{-1} + \int_2^{[X]} (\log u)^{-1} \mathrm{d}u$$

and

$$(\log 2)^{-1} + (\log 3)^{-1} + \cdots + (\log [X])^{-1} > \int_2^{[X]} (\log u)^{-1} \mathrm{d}u + (\log [X])^{-1}.$$

It follows that

$$(\log 2)^{-1} + (\log 3)^{-1} + \cdots + (\log [X])^{-1} = \int_2^{X} (\log u)^{-1} \mathrm{d}u + O(1).$$

The formula above for $\pi(X)$ combined with this and the estimate for $E(X)$ yields the theorem.

## 4.6

We now turn to the problem of studying the distribution of the zeros from another standpoint. Let

$$N(T) = \mathrm{Card}\,\{\rho \in Z : 0 < \mathrm{Im}(\rho) < T\}.$$

As was indicated in § 1.5 Riemann described without proof the asymptotic behaviour of this function and his claims were justified in 1895 by von Mangoldt. In order to describe the results it is convenient to introduce the function $S(T)$ which is defined for those $T > 0$ such that $T \notin \{\mathrm{Im}(\rho) : \rho \in Z\}$. It is defined as the variation of $\pi^{-1} \arg \zeta(s)$ along the path consisting of a segment from 2 to $2 + \mathrm{i}T$ and $2 + \mathrm{i}T$ to $1/2 + \mathrm{i}T$. The number 2 here could be replaced by any other $\sigma > 1$ without changing the value of the function.

## 4.7

The function $S$ is not very well understood. The estimate that we need is given in the following theorem.

**Theorem:** *One has as* $T \to \infty$ *that* $S(T) = O(\log T)$.

*Proof*: We have that

$$\begin{aligned} |\zeta(2 + \mathrm{i}t) - 1| &= \left| \sum_{n \geqslant 2} n^{-2-\mathrm{i}t} \right| \\ &\leqslant \sum_{n \geqslant 2} n^{-2} \\ &< 1. \end{aligned}$$

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Thus $\mathrm{Re}\{\zeta(2+it)\} > 0$ and hence the variation of $\arg\zeta$ along the segment joining 2 to $2+iT$ is at most $\pi/2$.

The variation along the segment joining $2+iT$ to $1/2+iT$ is the imaginary part of the integral

$$\int_{2+iT}^{1/2+iT} \zeta'(s)/\zeta(s)\,ds.$$

By Theorem 3.3 this is

$$\sum_{\substack{\rho\in Z \\ |\mathrm{Im}(\rho)-T|<1}} \mathrm{Im}\left\{\int_{2+iT}^{1/2+iT} (s-\rho)^{-1}\,ds\right\} + O(\log T).$$

In the sum here each summand is in absolute value at most $\pi$ and by Corollary 3.4 there are at most $O(\log T)$ summands. Hence the entire sum is $O(\log T)$, as asserted.

## 4.8

Davenport remarks that this is just about the only estimate that has not been improved in this century. It is known that $S$ oscillates quite violently – for example an application of Littlewood's Theorem (A7.1, see Exercise 4.9) yields

$$\int_0^T S(t)\,dt = O(\log T),$$

an estimate also due to Littlewood. If $S$ did not oscillate one would expect $O(T\log T)$ for this estimate. The numerical evidence indicates that $S(T)$ is very small; for $T < 545\,439\,823$ recent computations of te Riele, van de Lune and Winter (te Riele, van de Lune & Winter (1)) show that $S(T)$ does not exceed 2.32 and is 'usually' less than 2. At zeros it jumps by the multiplicity and hence by at least 1. Thus it cannot be estimated much more precisely than by 1. To date there are no known zeros of multiplicity greater than 1. These results might indicate that Theorem 4.7 is far from the truth. Indeed on the Lindelöf Hypothesis (§ 5.7) one can show that $S(t) = o(\log t)$ and on the (stronger) Riemann Hypothesis (§ 5.1) that $S(t) = O(\log|t|/\log\log|t|)$. In the opposite direction one can show, also under the Riemann Hypothesis, that for any $\varepsilon > 0$ there exists a sequence $t_m \to \infty$ such that

$$S(t_m) > (\log t_m)^{1/2-\varepsilon}$$

and a similar result for negative values. If the Riemann Hypothesis is not assumed then one can prove analogous results but with the exponent 1/2 replaced by 1/3. There are also similar statements for the integrated form

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$\int_0^T S(t)dt$. The numerical computations and these theorems are not at all at variance as $(\log t)^{1/2}$ grows very slowly – for example

$$(\log 545\,439\,823)^{1/2} = 4.485\,2\ldots$$

For more details on these questions see Titchmarsh (2) §§9.9–9.13 and 14.10–14.13 and Ghosh (1).

## 4.9

We now come to the description of the asymptotic behaviour of the function $N$.

**Theorem:** *One has, as* $T \to \infty$, $T \notin \{\mathrm{Im}(\rho): \rho \in Z\}$

$$N(T) = (T/2\pi)\log(T/2\pi) - (T/2\pi) + 7/8 + S(T) + O(T^{-1})$$

*and*

$$N(T) = (T/2\pi)\log(T/2\pi) - (T/2\pi) + O(\log T).$$

Before we prove this we note that the first estimate suffices, given the knowledge of $S(T)$, to compute $N(T)$ precisely for large enough $T$ since $N(T)$ has to be an integer. The techniques described in Chapter 2, and more often the 'approximate functional equation' suffice to compute $\zeta$ with sufficient accuracy to determine $S(T)$. On the other hand by means of the integral representation theorems such as that of §2.10, or by the 'approximate function equation', one can compute $\zeta$ on the critical line and determine the number of real zeros of $\zeta(1/2 + it)$ by real variable techniques. By comparing the number of zeros on the line $\{s: \mathrm{Re}(s) = 1/2\}$ with $N(T)$ one shows that there are no zeros which do not lie on the critical line. In this way, with various computational improvements, it has been verified that the first 1 500 000 001 zeros all lie on this line. They are simple which is what one would expect from the very small sizes of $S$ in the same range, but the difference between two consecutive zeros can be much smaller than the expected $2\pi/\log T$.

For more details on the computation of the Riemann zeta-function see Edwards (1), Titchmarsh (2); for tables of the zeros see Haselgrove (1), Haselgrove & Miller (1), Spira (1) and for the most extensive computations see te Riele *et al.* (1).

## 4.10

We shall now prove Theorem 4.9. It is clear that in view of Theorem 4.7 we need only prove the first estimate. We introduce first a notation which

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will be useful. Let $f$ be any complex valued function on an open set in $\mathbf{C}$. Let $x, y$ be two points at which $f$ is defined and such that $f$ is defined, continuous and nowhere zero on the segment joining $x$ to $y$. On this segment we can define a continuous branch of $\arg f$; we let

$$\Delta_x^y \arg f = (\arg f)(y) - (\arg f)(x).$$

This does not depend on the choice of the branch taken.

Let

$$\xi(s) = \pi^{-s/2}\Gamma(s/2)\zeta(s).$$

The zeros of this function in $\{s \in \mathbf{C}: -1 \leqslant \mathrm{Re}(s) \leqslant 2\}$ are those of $\zeta$ and it has two simple poles. Let $T$ be such that $T \notin \{\mathrm{Im}(\rho): \rho \in Z\}$. By the principle of the argument of complex analysis one has

$$2\pi\{2N(T) - 2\} = \Delta_{2-iT}^{2+iT} \arg \xi + \Delta_{2+iT}^{-1+iT} \arg \xi + \Delta_{-1+iT}^{-1-iT} \arg \xi + \Delta_{-1-iT}^{2-iT} \arg \xi.$$

Recall next that $\xi(s) = \xi(1-s)$ and $\overline{\xi}(s) = \xi(\bar{s})$. It follows that

$$\Delta_2^{2+iT} \arg \xi = \Delta_{2-iT}^{2} \arg \xi = -\Delta_{-1}^{-1+iT} \arg \xi = -\Delta_{-1-iT}^{-1} \arg \xi$$

and

$$\Delta_{2+iT}^{1/2+iT} \arg \xi = \Delta_{1/2+iT}^{-1+iT} \arg \xi = -\Delta_{2-iT}^{1/2-iT} \arg \xi = -\Delta_{1/2-iT}^{-1-iT} \arg \xi.$$

Consequently we have

$$2\pi\{2N(T) - 2\} = 4(\Delta_2^{2+iT} \arg \xi + \Delta_{2+iT}^{1/2+iT} \arg \xi).$$

We next use the fact that

$$\Delta_x^y \arg f_1 f_2 = \Delta_x^y \arg f_1 + \Delta_x^y \arg f_2$$

and obtain

$$N(T) = 1 + \pi^{-1}(\Delta_2^{2+iT} \arg \{\pi^{-s/2}\Gamma(s/2)\} + \Delta_{2+iT}^{1/2+iT} \arg \{\pi^{-s/2}\Gamma(s/2)\}) + \pi^{-1}(\Delta_2^{2+iT} \arg \zeta + \Delta_{2+iT}^{1/2+iT} \arg \zeta).$$

The third term is by definition $S(T)$. Since $\log \Gamma$ is defined on $\{s: \mathrm{Re}(s) > 0\}$ the second term is

$$\pi^{-1} \mathrm{Im}\{\log \Gamma(1/4 + iT/2)\} - (T/2\pi)\log \pi.$$

By the asymptotic expansion of $\log \Gamma$ (Stirling's Formula) we have

$$\mathrm{Im}\{\log \Gamma(1/4 + iT/2)\} = (T/2)\log(T/2) - T/2 - \pi/8 + O(T^{-1}).$$

Combining these we obtain the formula stated in the theorem.

## 4.11

In the following sections we shall give an estimate on the number of zeros which do not lie on the critical line $\{s \in \mathbf{C}: \mathrm{Re}(s) = 1/2\}$. We have already

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seen in Theorem 4.2 that the zeros of the zeta-function cannot lie too close to the line $\{s\in\mathbf{C}: \mathrm{Re}(s)=1\}$, but this does not rule out the possibility that many of the zeros lie, say, on the line $\{s\in\mathbf{C}: \mathrm{Re}(s)=3/4\}$. We shall prove, by a method due to Littlewood, a result of Bohr and Landau to the effect that almost all of the zeros lie close to the critical line. This proof also has the feature that it demonstrates the value of certain integral estimates of the zeta-function, and in particular Theorem 4.12 below. Such estimates as this are very closely related to the 'approximate functional equation' but we shall give a proof which is a little simpler, and instructive in its own right, but which is related in its techniques with the 'approximate functional equation'.

Define for $T>0$ and $\sigma>1/2$ the quantity

$$N(\sigma,T)=\mathrm{Card}\{\rho\in Z: \mathrm{Re}(\rho)>\sigma, 0<\mathrm{Im}(\rho)<T\}.$$

**Theorem**: *For $\sigma>1/2$ one has (for fixed $\sigma$) as $T\to\infty$*

$$N(\sigma,T)=O(T).$$

Observe that for $\sigma>1/2$

$$N(\sigma,T)=o(N(T))$$

which expresses the fact that for any $\sigma>1/2$ almost all of the zeros lie in $\{s\in\mathbf{C}: 1-\sigma<\mathrm{Re}(s)<\sigma\}$. The estimate given above is by no means the best that is known and a great deal of work has been done in improving this estimate. For a survey of this see Heath-Brown (2).

## 4.12

For the proof of Theorem 4.11 we shall need the following estimate which is important in its own right.

**Theorem:** *For $\sigma>1/2$, $\sigma$ fixed we have as $T\to\infty$*

$$\int_0^T |\zeta(\sigma+\mathrm{i}t)|^2\,\mathrm{d}t=O(T).$$

It is possible to replace this bound by an asymptotic formula, for example the $O(T)$-term can be replaced by $(\sigma<1)$ $\zeta(2\sigma)T+O(T^{2-2\sigma}\log T+T^{1/2}(\log T)^{1/2})$ (see Titchmarsh (2), § 7.2). We shall not go into this question and its variants further here.

## 4.13

We shall first deduce Theorem 4.11 from Theorem 4.12. The crucial technical tool here is an integrated form of the principle of the argument

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due to Littlewood, who first recognized its significance for such problems. This theorem is stated and proved in A7.1. Applied to our case it shows that for $\sigma_1 > 1/2$ and $T > 0$, $T \notin \mathrm{Im}(Z)$ and suitably small $\varepsilon > 0$

$$2\pi \int_{\sigma_1}^{1} N(\sigma, T)\mathrm{d}\sigma = \int_{\varepsilon}^{T} \{\log|\zeta(\sigma_1 + \mathrm{i}t)| - \log|\zeta(2 + \mathrm{i}t)|\}\,\mathrm{d}t + \int_{\sigma_1}^{2} \{\arg\zeta(v + \mathrm{i}T) - \arg\zeta(v + \mathrm{i}\varepsilon)\}\,\mathrm{d}v$$

where $\arg\zeta$ is determined by continuous variation along the sides of the rectangle with corners $2 + \mathrm{i}\varepsilon$, $2 + \mathrm{i}T$, $\sigma_1 + \mathrm{i}T, \sigma_1 + \mathrm{i}\varepsilon$. We have already seen that $\arg\zeta(v + \mathrm{i}T) = O(\log T)$. Clearly $\arg\zeta(v + \mathrm{i}\varepsilon) = O(1)$ and $\log|\zeta(2 + \mathrm{i}t)| = O(1)$. Thus we see that

$$2\pi \int_{\sigma_1}^{1} N(\sigma, T)\mathrm{d}\sigma = \int_{\varepsilon}^{T} \log|\zeta(\sigma_1 + \mathrm{i}t)|\mathrm{d}t + O(T).$$

By Jensen's inequality, in this case the integral form of the geometric mean–arithmetic mean inequality,

$$T^{-1} \int_{\varepsilon}^{T} \log|\zeta(\sigma_1 + \mathrm{i}t)|^2 \mathrm{d}t \leqslant \log\left\{T^{-1} \int_{\varepsilon}^{T} |\zeta(\sigma_1 + \mathrm{i}t)|^2 \mathrm{d}t\right\}.$$

By Theorem 4.12 the right-hand side is bounded which shows that

$$2\pi \int_{\sigma_1}^{1} N(\sigma, T)\mathrm{d}\sigma = O(T).$$

Choose now for a given $\sigma_2$ with $1/2 < \sigma_2 < 1$ a $\sigma_1$ with $1/2 < \sigma_1 < \sigma_2$. For $\sigma$ with $\sigma_1 < \sigma < \sigma_2$ we have

$$N(\sigma, T) > N(\sigma_2, T)$$

and hence

$$2\pi(\sigma_2 - \sigma_1)N(\sigma_2, T) \leqslant 2\pi \int_{\sigma_1}^{1} N(\sigma, T)\mathrm{d}\sigma.$$

From this it follows that

$$N(\sigma_2, T) = O(T)$$

which is what we had to prove.

## 4.14

It remains to prove Theorem 4.12. To do this we start from the inverse Mellin transform of the gamma-function

$$(1/2\pi\mathrm{i}) \int_{(\sigma)} \Gamma(s)x^{-s}\mathrm{d}s = \mathrm{e}^{-x}$$

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which is valid for $\sigma > 0$. We suppose now that $\sigma > 1$ and replace $x$ by $nx$ and sum; this gives the formula

$$(1/2\pi i)\int_{(\sigma)} \Gamma(s)\zeta(s)x^{-s}ds = (e^x - 1)^{-1}$$

which is the Mellin inverse transform of one of the formulae used to prove the analytic continuation of the zeta-function. In view of Stirling's Formula the convergence is absolute. For the same reason we can move the line of integration to $(\sigma)$ with $1/2 < \sigma < 1$. The pole of the zeta-function at $s = 1$ yields a residue $x^{-1}$ and so we have

$$(1/2\pi i)\int_{(\sigma)} \Gamma(s)\zeta(s)x^{-s}ds = (e^x - 1)^{-1} - x^{-1}.$$

We note next that this formula is meaningful when $\mathrm{Re}(x) > 0$ and $x^{-s} = e^{\log x}$ with $-\pi < \mathrm{Im}(\log x) < \pi$. As both sides are analytic in this region the identity persists by the principle of analytic continuation. Thus we can use this identity on $e^{i(\pi/2-\delta)}\mathbf{R}_+^\times$ where $0 < \delta < \pi/2$. Let $z = e^{i(\pi/2-\delta)}$; then we obtain

$$(1/2\pi i)\int_{(\sigma)} \Gamma(s)\zeta(s)e^{-is(\pi/2-\delta)}x^{-s}ds = (e^{zx} - 1)^{-1} - (zx)^{-1}.$$

We write this as

$$(2\pi)^{-1}\int_{-\infty}^{\infty} \Gamma(\sigma + it)\zeta(\sigma + it)e^{-i(\sigma+it)(\pi/2-\delta)}e^{-it\log x}dt$$
$$= \{(e^{zx} - 1)^{-1} - (zx)^{-1}\}x^{\sigma}$$

which we recognize as a Fourier transform. We can apply Parseval's Formula (A1.10) as both sides are square-integrable and obtain

$$(2\pi)^{-1}\int_{-\infty}^{\infty} |\Gamma(\sigma + it)\zeta(\sigma + it)|^2 e^{t(\pi-2\delta)}dt$$
$$= \int_0^{\infty} x^{2\sigma-1}|(e^{xz} - 1)^{-1} - (zx)^{-1}|^2 dx.$$

We shall now study the behaviour of both sides of this formula as $\delta \to 0$, and thus $z \to i$. The crux of the argument is to show that the right-hand side $\sim \zeta(2\sigma)(2\delta)^{-2\sigma}\Gamma(2\sigma)$ as $\delta \to 0$. It is this part of the argument that is close to those used in the proof of the 'approximate functional equation'.

To prove this we observe first that the part of the second integral taken over $]0, \pi[$ is continuous in $\delta$ in a neighbourhood of 0, and is therefore bounded. The integral over $[\pi, \infty[$ is when multiplied out a sum of four terms

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$$\int_\pi^\infty x^{2\sigma-1}(e^{zx}-1)^{-1}(e^{z^{-1}x}-1)^{-1}\,dx - z\int_\pi^\infty x^{2\sigma-1}(e^{zx}-1)^{-1}x^{-1}\,dx$$
$$- z^{-1}\int_\pi^\infty x^{2\sigma-1}(e^{z^{-1}x}-1)^{-1}x^{-1}\,dx + \int_\pi^\infty x^{2\sigma-2}x^{-2}\,dx.$$

The last term does not depend on $\delta$ and is therefore also bounded.

The first term is

$$\sum_{m,n\geqslant 1}\int_\pi^\infty e^{-(mz+nz^{-1})x}x^{2\sigma-1}\,dx.$$

Let us consider first those terms with $m=n$. As $z+z^{-1}=2\sin\delta$ these are

$$\sum_m \int_\pi^\infty e^{-2mx\sin\delta}x^{2\sigma-1}\,dx = \int_\pi^\infty (e^{2x\sin\delta}-1)^{-1}x^{2\sigma-1}\,dx.$$

After a change of variable of $x$ to $x/(2\sin\delta)$ this becomes

$$\int_{2\pi\sin\delta}^\infty (e^x-1)^{-1}x^{2\sigma-1}\,dx(2\sin\delta)^{-2\sigma}.$$

As $\delta\to 0$ the integral converges to

$$\zeta(2\sigma)\Gamma(2\sigma).$$

Thus this part is asymptotic to $\zeta(2\sigma)\Gamma(2\sigma)(2\delta)^{-2\sigma}$ which is the main term. We have now to show that the remaining terms are of a smaller order of magnitude as $\delta\to 0$. Consider those terms in the integral with $m\neq n$. For definiteness we shall study those with $m<n$; the case of $m>n$ is similar. We have to estimate the expression

$$\sum_{m<n}\int_\pi^\infty e^{-\{(m+n)\sin\delta+(m-n)i\cos\delta\}x}x^{2\sigma-1}\,dx.$$

To the integral we apply Cauchy's Theorem and replace the integral over $]\pi,\infty[$ by the line $y\mapsto\pi+iy, y>0$. It is easy to justify this. On this line the absolute value of the integrand is

$$e^{-\pi(m+n)\sin\delta-|m-n|y\cos\delta}(\pi^2+y^2)^{\sigma-1/2}$$

so that the sum is dominated by

$$\sum_{m<n} O(|m-n|^{-1}e^{-\pi(m+n)\sin\delta})$$

since $|m-n|\cos\delta$ is bounded below and for $A>1$ one has by a change of variables the following inequality for $u>0$

$$\int_0^\infty e^{-Ax}(x^2+\pi^2)^u dx \leqslant A^{-1}\int_0^\infty e^{-x}(x^2+\pi^2)^u dx.$$

In the estimate that we obtained above let $l=n-m$ so that the variable is

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restricted by $l < n$. Thus we have the estimate (on making use of $m + n > n$)

$$\sum_n \sum_{l<n} O(l^{-1} e^{-\pi n \sin\delta}).$$

The sum over $l$ is $O(\log n)$. To estimate the resulting sum

$$\sum_{n \geqslant 2} \log n e^{-\pi n \sin\delta}$$

it is most convenient to split the summation into two parts, that where $n \leqslant \delta^{-1}$ and that where $n > \delta^{-1}$. The first of these consists of $O(\delta^{-1})$ terms each of which is $O(\log \delta^{-1})$. The estimate that we thus obtain is $O(\delta^{-1} \log \delta^{-1})$. The remaining sum can be written as

$$\log \delta^{-1} \sum_{n\delta > 1} e^{-\pi n \sin\delta} + \sum_{n\delta > 1} \log n\delta \, e^{-\pi n \sin\delta}.$$

The first of the sums is a geometric series which can be evaluated and this yields an estimate of again $O(\delta^{-1} \log \delta^{-1})$. Finally $\log(\eta\delta)e^{-\pi n \sin\delta} = O(e^{-n\delta/2})$. Thus the second term can be estimated by $O(\delta^{-1})$. Hence the first of our four integrals is asymptotic to $\zeta(2\sigma)\Gamma(2\delta)(2\delta)^{-2\sigma}$.

It remains to examine the second and third integrals. They are complex conjugates of one another and so we need only estimate one of them. By an analogous application of Cauchy's Theorem to that used above we see that

$$\int_\pi^\infty x^{2\sigma-1}(e^{xz} - 1)^{-1} dx = i \int_\pi^\infty (\pi + iy)^{2\sigma-1}(e^{z(\pi+iy)} - 1)^{-1} dy.$$

For $\delta < \pi/4$ we have

$$(e^{z(\pi+iy)} - 1)^{-1} = O(e^{-y/\sqrt{2}})$$

from which we see that this integral is bounded as $\delta \to 0$.

We have now completed the proof that as $\delta \to 0$

$$\int_{-\infty}^\infty |\Gamma(\sigma + it)\zeta(\sigma + it)|^2 e^{t(\pi - 2\delta)} dt \sim \zeta(2\sigma)\Gamma(2\sigma)(2\delta)^{-2\sigma}.$$

We can now deduce Theorem 4.12. From the asymptotic formula we have for $T > 1$

$$\int_{T/2}^T |\Gamma(\sigma + it)\zeta(\sigma + it)|^2 e^{t(\pi - 2\delta)} dt = O(\delta^{-2\sigma}).$$

In this we take $\delta = T^{-1}$ and recall from Stirling's Formula that

$$|\Gamma(\sigma + it)|^2 e^{\pi t} \sim \pi |t|^{2\sigma - 1}.$$

Thus we find that

$$\int_{T/2}^T |\zeta(\sigma + it)|^2 e^{-2t/T} t^{2\sigma-1} dt = O(T^{2\sigma}),$$

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or, more simply

$$\int_{T/2}^{T} |\zeta(\sigma + it)|^2 \, dt = O(T).$$

We now replace $T$ by $T/2$, $T/4$, etc. and sum the ensuing inequalities which yields Theorem 4.12.

## 4.15

We remark that the estimate of Theorem 4.12 can be replaced by an asymptotic formula

$$\int_{0}^{T} |\zeta(\sigma + it)|^2 \, dt \sim T\zeta(2\sigma).$$

The leading term can be obtained by sharpening the argument given above through the use of a Tauberian theorem. A similar asymptotic formula is known for $\int_0^T |\zeta(\sigma + it)|^4 \, dt$ (see Titchmarsh (2), §7.5); beyond this the only non-trivial estimate available for higher powers, and not deducible from the results already quoted, is Heath-Brown's estimate

$$\int_{0}^{T} |\zeta(1/2 + it)|^{12} \, dt = O(T^2 (\log T)^{17}).$$

For further details of this and related questions the reader should consult Heath-Brown (1) and also Titchmarsh (2), Chapter 7.

## Exercises

4.1 For $c > 0$ let $H_c = \{s \in \mathbf{C} : |1 - c/\log|\mathrm{Im}(s)|| < \mathrm{Re}(s) < 2, |\mathrm{Im}(s)| > 2\}$. Show that for suitable $c$ one has for $s \in H_c$

$$\zeta(s) = O(\log |s|).$$

4.2 In the notation of Exercise 4.1 show that for suitable $c$ one has for $s \in H_c$

$$\zeta'(s) = O((\log |s|)^2)$$

by adapting the argument of Exercise 4.1. Deduce it also from Exercise 4.1.

4.3 Use the argument of §4.2 to show that as $|t| \to \infty$

$$\zeta(1 + it)^{-1} = O((\log |t|)^7)$$
$$\zeta'/\zeta(1 + it) = O((\log |t|)^9).$$

4.4 Suppose that $a, b$ are increasing functions such that $a \geqslant 1$, $b > 0$, $ab = o(e^b)$ and $\zeta(s) = O(e^{b(|\mathrm{Im}(s)|)})$ in $\{s \in \mathbf{C} : \mathrm{Re}(s) \geqslant 1 - a(|\mathrm{Im}(s)|)^{-1}\}$. Show that there exists $c > 0$ such that if $\rho \in Z$ then

$$\mathrm{Re}(\rho) < 1 - cab(2|\mathrm{Im}(\rho)| + 1)^{-1}.$$

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Hint: use Theorem A6.2; the method of §4.2, yields a much weaker result. This statement apparently represents the best one can achieve using Hadamard's inequality. Note that zero-free strips of the form $\{s \in \mathbf{C}: \mathrm{Re}(s) > \eta\}$, $\eta < 1$ are not within the range of this method.

4.5 Under the assumptions of Exercise 4.4 show that there exists $c' > 0$ so that as $|s| \to \infty$, $1 - c'ab\{2|\mathrm{Im}(s)| + 3\}^{-1} < \mathrm{Re}(s) < 2$ one has

$$\zeta'(s)/\zeta(s) = O((ab)(2|s| + 3))$$
$$\zeta(s)^{-1} = O((ab)(2|s| + 3)).$$

4.6 Deduce from Exercise 4.5 that for a suitable $c > 0$ then as $|s| \to \infty$ with $1 - c/\log\{|\mathrm{Im}(s)| + 2\} < \mathrm{Re}(s) < 2$ one has

$$\zeta'(s)/\zeta(s) = O(\log|s|)$$
$$\zeta(s)^{-1} = O(\log|s|).$$

4.7 Using new bounds for exponential sums and the approximate functional equation Vinogradov proved that there exist $c, c' > 0$ such that as $|s| \to \infty$ with

$$1 - c\{\log\log|\mathrm{Im}(s)|\}^{1/2}/\{\log|\mathrm{Im}(s)|\}^{1/2} < \mathrm{Re}(s) < 2$$

one has

$$\zeta(s) = O(\exp\{c'(\log|s|)^{1/4}(\log\log|s|)^{1/2}/(\log\log|s|)^{1/2}\}).$$

What zero-free region for $\zeta$ does this imply? What bound on $\zeta(1 + it)^{-1}$ as $|t| \to \infty$ can one deduce? What are the consequences for the error term in the Prime Number Theorem?

4.8 Show that $S(t)$ can be defined as $\pi^{-1}\Delta_{\infty + it}^{1/2 + it}\arg(\zeta)$.

4.9 Use Exercise 4.8 and the argument of §4.13 to show that

$$\int_0^T S(t)\mathrm{d}t = O(\log T) \qquad \text{as } T \to \infty.$$

4.10 Let $(a_n)$ be a sequence and $A$ a positive non-decreasing function so that $|a_n| \leqslant A(n)$. Suppose that $\sum_{n \geqslant 1} a_n n^{-s}$ converges absolutely if $\mathrm{Re}(s) > 1$ and that there exists $\alpha \in \mathbf{R}_+$ so that as $s \to 1, s > 1$

$$\sum_{n \geqslant 1} |a_n| n^{-s} = O((s-1)^{-\alpha}).$$

Let $c > 0$; show that if $\mathrm{Re}(s) > 1 - c$ and $X \in \mathbf{R}_+ - \mathbf{N}$ then

$$\sum_{n \geqslant X} a_n n^{-s} = (2\pi\mathrm{i})^{-1} \int_{c - \mathrm{i}T}^{c + \mathrm{i}T} \left( \sum_{m \geqslant 1} a_m m^{-(s+t)} \right) X^t t^{-1} \mathrm{d}t$$
$$+ O(X^c T^{-1}\{\mathrm{Re}(s) + c - 1\}^{-\alpha}) + O(A(2X)X^{1 - \mathrm{Re}(s)}\log X)$$
$$+ O(A(X+1)X^{1-\mathrm{Re}(s)}T^{-1}\langle X\rangle^{-1}),$$

where $\langle X\rangle = \mathrm{Min}\{|X - m| : m \in \mathbf{Z}\}$.

4.11 Use Exercise 4.10 to prove Theorem 4.3.

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4.12 Show that for any $\varepsilon > 0$

$$\sum_{n \geqslant X} \mathrm{d}(n) = X \operatorname*{Res}_{t=1} \{\zeta(t)^2 X^{t-1}/t\} + O(X^{1/2+\varepsilon})$$

$$\sum_{n \leqslant X} \mathrm{d}(n)^2 = X \operatorname*{Res}_{t=1} \{\zeta(t)^4 X^{t-1}/\zeta(2t)t\} + O(X^{3/4+\varepsilon})$$

where $d$ is the divisor function ($d = d_2$).

4.13 Show that for any $m \geqslant 1$ that there exists a polynomial $P_m$ of degree $2^{m-1}$ so that

$$\sum_{n \leqslant X} \mathrm{d}(n)^m = X P_m(\log X) + O(X^{1-2^{-m}+\varepsilon})$$

for any $\varepsilon > 0$. Deduce that for no $M > 0$ an estimate $\mathrm{d}(n) = O((\log n)^M)$ can hold.

4.14 Find a sequence $n_i \to \infty$ so that for any $\eta > 0$ there exists $i_0(n)$ so that for $i > i_0(\eta)$

$$\mathrm{d}(n_i) > \exp\{(1-\eta)(\log 2)(\log n_i)/\log\log n_i\}.$$

By sharpening the argument of Exercise 1.5 show that for any $\eta > 0$ one has

$$\mathrm{d}(n_i) = O(\exp\{(1+\eta)(\log 2)(\log n_i)/\log\log n_i\}).$$

4.15 Show that for any $\varepsilon > 0$

$$\mathrm{Card}\{n | \phi(n) \leqslant X\} = CX + O(X^{7/9+\varepsilon})$$

where

$$C = \prod_p (1 + \{p(p-1)\}^{-1}).$$

(Hint: recall Exercise 1.16.)

4.16 Estimate the 'constant', as a function of $\sigma$, in Theorem 4.11.

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